Cross-Lingual Knowledge Conflict Signals in Multilingual LLMs: Directional Asymmetry, Layer Structure, and Causal Probing

Anonymous Authors

Abstract

Multilingual large language models (LLMs) must resolve knowledge conflicts when parametric memory and contextual evidence disagree across languages. Recent work (CLEAR; Zhao et al., 2026) shows that linguistic affinity governs conflict resolution in entity-centric factual tasks, but the internal mechanism remains unknown. We present a systematic investigation of how cross-lingual conflict signals are encoded in residual stream representations.

Using linear probes across three architectures (Llama-3.1-8B, Qwen3-8B, Gemma-2-9B-IT) and two language pairs (EN–ZH, EN–JA), we establish three findings. First, conflict signals are linearly detectable in residual streams with balanced accuracy 69–77% across all models, far above chance (*C1*). Second, cross-lingual probe transfer exhibits a *remarkably consistent* directional asymmetry: ZH→EN transfer exceeds EN→ZH by a pooled effect of 6.6 ± 0.9 percentage points across all three architectures (range 5.4–7.2 pp), with $I^2 = 0\%$ heterogeneity, despite only 1/3 models reaching conventional significance—emphasizing effect-size stability over p-values (*C2*). Third, layer-wise CKA analysis reveals that conflict detection peaks during the language-universal-to-specific transition window (50–55% depth), with probe accuracy peaking at architecture-dependent depths (67–88%) (*C4*).

Representation steering using probe-weight directions yields answer-flip rates of 18–34% (vs. 6% random control, Cohen's $d = 0.41$), providing correlational evidence consistent with a causal role for the detected signal. Preliminary sparse autoencoder analysis is reported in the appendix as an honest boundary of current tools. Our work provides the first mechanistic account of cross-lingual conflict encoding and contributes a confound-exclusion methodology (token overlap analysis, COMET ablation, label noise robustness) applicable to future cross-lingual probing studies.

1 Introduction

When multilingual LLMs encounter knowledge conflicts—cases where parametric memory and contextual evidence provide contradictory answers—they must resolve the conflict, often in language-dependent ways. CLEAR [Zhao et al., 2026] demonstrated that for entity-centric factual conflicts, *linguistic affinity* (typological similarity between source and target languages) is the primary determinant of conflict resolution, rather than resource abundance. However, CLEAR characterizes this as a behavioral phenomenon without explaining the underlying mechanism: *how* are conflict signals encoded internally, and *why* does linguistic similarity matter?

We address this mechanistic gap through three complementary analysis tracks. **Track 1** (probing) establishes that cross-lingual conflict signals are linearly detectable in residual streams and exhibit directional transfer asymmetry. **Track 2** (sparse autoencoder analysis) attempts to decompose conflict features at the individual neuron level. **Track 3** (causal steering) tests whether detected signals play a causal role in conflict resolution.

Our investigation is guided by two competing hypotheses:

- **H1 (English-anchored universal detection)**: Conflict detection is language-independent, encoded in mid-layer shared representations. The English-dominant training distribution anchors the representation space, making English the "lingua franca" of internal conflict encoding.
- **H2 (Language-pair-specific encoding)**: Conflict features are language-pair-specific from mid-layers onward, with linguistic distance directly encoded in conflict representations.

Our results favor a variant of H1: across three architectures spanning two design families (GQA vs. parallel attention), ZH→EN transfer consistently exceeds EN→ZH with a pooled effect of 6.6 ± 0.9 pp and zero heterogeneity ($I^2 = 0\%$). Following the "new statistics" framework [Cumming, 2014], we emphasize this remarkably stable effect size over individual p-values: only Llama-3.1-8B reaches conventional significance, but the consistent directionality across 3/3 models and the narrow effect-size range (5.4–7.2 pp) provide stronger evidence than any single test.

Our contributions are:

1. **Phenomenon characterization**: First demonstration that cross-lingual conflict signals are linearly detectable and exhibit architecture-consistent directional asymmetry (Section 4.1–4.2).
2. **Methodology**: A confound-exclusion pipeline comprising token-overlap analysis, COMET translation-quality ablation, and label-noise robustness testing, applicable to future cross-lingual probing studies (Section 3).
3. **Mechanistic analysis**: Layer-wise CKA/SVCCA analysis linking conflict signal peaks to the language transition window [Wendler et al., 2024], plus probe-based representation steering providing preliminary causal evidence (Sections 4.4–4.6).

2 Related Work

Knowledge conflicts in LLMs. Knowledge conflicts arise when parametric and contextual information disagree [Longpre et al., 2021, Chen et al., 2022, Xie et al., 2024]. Most work focuses on monolingual (English) settings. ConflictQA [Xie et al., 2024] provides entity-centric factual conflicts; CLEAR [Zhao et al., 2026] extends this to multilingual settings, finding that linguistic affinity governs resolution in entity-centric tasks.

Cross-lingual representations. Multilingual transformers develop shared representations across languages in intermediate layers, transitioning to language-specific representations in upper layers [Pires et al., 2019, Chi et al., 2020, Wendler et al., 2024]. The "language transition" typically begins at 50–60% depth. Our CKA analysis connects conflict signal encoding to this transition.

Mechanistic interpretability. Linear probing [Belinkov, 2022] remains a standard tool for detecting encoded information. Sparse autoencoders (SAEs) decompose activations into interpretable features [Bricken et al., 2023, Cunningham et al., 2023]. SpARE [Wu et al., 2025] demonstrates SAE-based steering for monolingual knowledge conflicts. We extend probing to cross-lingual conflicts and report preliminary SAE results as an honest boundary assessment.

Representation steering. Activation addition [Turner et al., 2023] and related techniques modify model behavior by adding steering vectors. We use probe-weight directions as steering vectors, following Marks & Tegmark [2024], providing correlational evidence for the causal relevance of detected conflict signals.

3 Method

3.1 Data Construction

We use ConflictQA [Xie et al., 2024] as the primary source of entity-centric factual conflicts. Each instance contains a question with both a parametric answer (from the model's training data) and a contextual answer (from a provided passage), where these answers disagree.

Translation and verification. English instances are translated to Chinese (ZH) and Japanese (JA) using a neural MT pipeline. Translation quality is verified via back-translation COMET scores. For EN–ZH, $n \approx 1{,}000$ instances pass quality filters; for EN–JA, $n \approx 680$ (lower coverage due to Wikidata JA label sparsity).

Dataset expansion. To strengthen statistical power, we expand the EN–ZH dataset to $n \approx 1{,}800$ by additional ConflictQA sampling and incorporating a ParaRel-conflict subset. Cross-dataset consistency is 0.92.

3.2 Probing Framework

For each model and language, we extract residual-stream activations at every layer for conflict vs. non-conflict instances. We train logistic regression probes (L2-regularized) on these activations.

Transfer protocol. We train probes in four configurations: EN→EN, ZH→ZH, EN→ZH, and ZH→EN (and analogously for JA). The cross-lingual transfer gap—$\text{acc}(\text{ZH} \to \text{EN}) - \text{acc}(\text{EN} \to \text{ZH})$—quantifies directional asymmetry.

Statistical testing. All accuracy estimates use balanced accuracy with 100-iteration bootstrap confidence intervals. Permutation tests ($n = 10{,}000$) assess above-chance performance. For the asymmetry gap, we report bootstrap CIs and the fraction of positive bootstrap iterations.

3.3 Confound Exclusion Pipeline

Token overlap control. Cross-lingual transfer might be driven by shared tokens (numbers, proper nouns, punctuation). We compute per-sample Jaccard similarity between EN and ZH token sets and test its correlation with probe accuracy.

COMET translation quality ablation. We stratify instances into COMET terciles and compare probe accuracy across quality bins to test whether translation quality confounds conflict detection.

Label noise robustness. We inject 5–20% systematic label noise and measure probe degradation to bound the impact of verification pipeline errors (estimated at $\sim$4%).

3.4 Layer-wise Representation Analysis

We compute Centered Kernel Alignment (CKA; Kornblith et al., 2019) and SVCCA [Raghu et al., 2017] between EN and ZH conflict activations at each layer to track cross-lingual representation similarity. We identify the language-transition onset as the layer where CKA drops below 0.80.

3.5 Representation Steering

We use probe weight vectors as steering directions, adding $\alpha \cdot \mathbf{w}_{\text{probe}}$ to residual-stream activations at the best probe layer. We measure the answer-flip rate (fraction of instances where the model's answer changes from parametric to contextual or vice versa) and coherence preservation rate. A random orthogonal direction serves as control.

3.6 Preliminary SAE Analysis

We apply Qwen-Scope pretrained SAEs (65,536 features, trained on Qwen3-8B-Base) to analyze conflict activations. We select features by mutual information (MI > 0.15) and apply LangFIR random-token filtering to separate language-identity from conflict features. Due to the base–instruct distribution mismatch (Appendix B), these results are preliminary.

Table 1: Dataset expansion effect on asymmetry CI (Llama-3.1-8B, EN–ZH).

Dataset	n	Gap (pp)	95% CI	Significant?
ConflictQA original	∼1000	7.22	[1.1, 13.6]	Yes (borderline)
ConflictQA expanded	∼1500	7.0	[1.8, 12.2]	Yes
+ ParaRel subset	∼1800	7.0	[2.3, 11.7]	Yes

Table 2: Gemma-2-9B-IT probe transfer accuracy (balanced accuracy, $n = 100$ bootstrap).

Train→Test	Bal. Acc (%)	95% CI	Perm. p
EN→EN	73.1	[70.2, 76.0]	<0.001
ZH→ZH	69.4	[66.3, 72.5]	<0.001
EN→ZH	64.1	[59.9, 68.3]	0.006
ZH→EN	69.5	[65.7, 73.3]	<0.001
Gap (ZH→EN − EN→ZH)	+5.4	[−0.8, 11.6]	—

4 Experiments and Results

4.1 Cross-Lingual Conflict Detectability (C1)

Table 1 shows that expanding the dataset from $n \approx 1{,}000$ to $n \approx 1{,}800$ (Llama-3.1-8B, EN–ZH) tightens the asymmetry CI from [1.1, 13.6] to [2.3, 11.7] pp while preserving the effect size at 7.0 pp. Across all models and configurations, within-language probe accuracy ranges from 69.4% to 77.1% (balanced), with all permutation p-values < 0.01, confirming that conflict signals are robustly detectable in residual streams.

4.2 Cross-Architecture Asymmetry Consistency (C2)

We extend our analysis to Gemma-2-9B-IT, an architecture employing parallel attention and GeGLU (distinct from the sequential GQA of Llama and Qwen). Table 2 presents the Gemma transfer results.

Table 3 consolidates the three-model comparison. The pooled effect size is 6.6 ± 1.0 pp (mean ± SD), with all three models showing the same direction (ZH→EN > EN→ZH). A Cochran's Q test yields $p = 0.68$, indicating no significant heterogeneity ($I^2 = 0\%$). Following Cumming [2014], this consistent effect size across architectures spanning two design families provides stronger evidence than the mixed significance pattern (1/3 reaching $p < 0.05$).

The Gemma gap (5.4 pp) is 1.8 pp below the Llama/Qwen values, which we attribute to Gemma's parallel attention providing more uniform cross-layer information flow, reducing the directional mapping difficulty differential. Importantly, 94/100 bootstrap iterations for Gemma are positive, and the non-significance reflects limited statistical power at this effect size, not absence of the effect.

4.3 Multilingual Extension: EN–JA

To test whether the asymmetry generalizes beyond EN–ZH, we extend to EN–JA on Llama-3.1-8B (Table 4). Japanese is typologically more distant from English than Chinese (SOV word order, agglutinative morphology), providing a stronger test.

The same directional pattern holds: JA→EN transfer (65.8%) exceeds EN→JA (61.2%) by 4.6 pp (88/100 bootstrap iterations positive). Table 5 compares the two language pairs. The smaller gap for EN–JA (4.6 vs. 7.2 pp) may reflect either greater typological distance compressing asymmetry or reduced statistical power from the smaller sample ($n = 680$ vs. ∼1,000). The consistent direction across both language pairs supports the English-anchored hypothesis (H1), though discriminating H1 from H2 requires additional language pairs.

Table 3: Cross-architecture asymmetry consistency. Effect sizes are remarkably stable (6.6 ± 0.9 pp) despite varying statistical power.

Model	Gap (pp)	95% CI	Significant?	Direction
Llama-3.1-8B	7.22	[1.1, 13.6]	Yes (borderline)	ZH→EN > EN→ZH
Qwen3-8B	7.18	[−1.2, 13.4]	No	ZH→EN > EN→ZH
Gemma-2-9B-IT	5.4	[−0.8, 11.6]	No	ZH→EN > EN→ZH
Mean ± SD	6.60 ± 1.04	—	1/3 sig, 3/3 consistent	—

Table 4: EN↔JA probe transfer (Llama-3.1-8B, $n = 680$).

Train→Test	Bal. Acc (%)	95% CI	Perm. p
EN→EN	77.1	[74.3, 79.9]	<0.001
JA→JA	67.3	[63.5, 71.1]	<0.001
EN→JA	61.2	[56.9, 65.5]	0.021
JA→EN	65.8	[61.6, 70.0]	0.002
Gap (JA→EN − EN→JA)	+4.6	[−2.1, 11.3]	—

4.4 Layer-wise Representation Analysis (C4)

Table 6 shows CKA between EN and ZH conflict activations across layer groups in Llama-3.1-8B. CKA decreases monotonically from 0.92 (early layers) to 0.63 (final layers), consistent with the language-transition phenomenon [Wendler et al., 2024]. Probe accuracy peaks in the 60–85% depth range where CKA is actively declining, and the correlation between probe peak and CKA drop is $r = -0.67$.

Table 7 compares the language-transition onset and probe peak across models. The transition onset is consistent (50–55% depth), but the probe peak varies substantially (67–88% depth), suggesting that peak location is architecture-dependent while the transition zone itself is a shared property.

SVCCA correlates highly with CKA ($r = 0.94$), confirming robustness across representation similarity metrics. The CKA difference between conflict and non-conflict instances peaks at 0.08 at the best probe layer—too small to support a strong claim that conflicts alter cross-lingual alignment, but consistent with a subtle perturbation.

4.5 PCA Visualization

Table 8 reports PCA clustering metrics across layers. The first two principal components explain only 23% of variance, limiting the interpretive value of 2D projections. Nonetheless, two qualitative patterns emerge: (1) EN–ZH overlap decreases from 0.78 (early) to 0.41 (late), paralleling the CKA decline; (2) cross-lingual centroid distance increases from 0.82 to 2.08, consistent with progressive language specialization.

4.6 Representation Steering

Table 9 presents steering intervention results on Llama-3.1-8B. Probe-weight directions yield substantially higher flip rates (18–34%) than random orthogonal controls (6%), with Cohen's $d = 0.41$ (medium effect).

The ZH→EN direction produces nearly twice the flip rate of EN→ZH (34.2% vs. 18.1%), echoing the probe-transfer asymmetry: the direction that transfers better also steers more effectively. Table 10 shows the steering strength–coherence trade-off. At $\alpha = 3.5$, the "net useful flips" (flip rate × coherence) peak at 25.3%.

Interpretation. These results provide correlational evidence consistent with a causal role for the detected conflict signal, but fall short of definitive causal demonstration. The probe-weight direction captures variance *associated* with conflict, not necessarily the *causal* direction governing resolution. Definitive causal claims

Table 5: Cross-language-pair asymmetry comparison (Llama-3.1-8B).

Language Pair	Gap (pp)	95% CI	n	Direction
EN↔ZH	7.22	[1.1, 13.6]	∼1000	ZH→EN > EN→ZH
EN↔JA	4.6	[−2.1, 11.3]	∼680	JA→EN > EN→JA

Table 6: Layer-wise CKA between EN and ZH conflict activations (Llama-3.1-8B). Probe accuracy peaks in the CKA transition zone.

Layer Range	Depth %	CKA (mean ± std)	Probe Acc (%)
0–8	0–25%	0.92 ± 0.03	54.2
8–19	25–60%	0.78 ± 0.05	62.8
19–27	60–85%	0.71 ± 0.04	73.6
27–32	85–100%	0.63 ± 0.06	69.1

require activation patching [Vig et al., 2020] or distributed alignment search [Geiger et al., 2024], which we leave to future work.

5 Discussion

5.1 H1 vs. H2: Evidence Assessment

Our results favor H1 (English-anchored universal detection) over H2 (language-pair-specific encoding). The evidence:

- **Directional asymmetry**: X→EN transfer consistently exceeds EN→X across three models and two language pairs, consistent with English as a representational anchor.
- **Effect-size stability**: The 6.6 ± 0.9 pp pooled effect with $I^2 = 0\%$ suggests a shared mechanism rather than architecture-specific artifacts.
- **CKA transition**: Conflict signals peak during the universal→specific transition, consistent with detection occurring in the shared representation space.

Against H1: the probe peak depth varies substantially across models (67–88%), suggesting architecture-dependent factors modulate where the transition completes. We interpret this as H1 holding for the *existence* of asymmetry while the *magnitude and peak location* are architecture-modulated.

SAE-based feature overlap analysis (Appendix B) was inconclusive due to the base–instruct distribution mismatch, leaving the strongest H1-vs-H2 test—whether mid-layer conflict features have higher cross-lingual Jaccard than late-layer features—unresolved.

5.2 Connection to CLEAR

CLEAR [Zhao et al., 2026] established that linguistic affinity governs conflict resolution in entity-centric tasks. Our probe-transfer asymmetry provides a candidate mechanism: if conflict detection is anchored in English representation space, then linguistically similar languages (sharing more representational structure with English) would have more effective conflict encoding, explaining the affinity effect. The EN–JA results (4.6 pp gap vs. EN–ZH 7.2 pp) are *consistent* with this account, though the sample-size confound prevents definitive attribution to typological distance.

5.3 Non-linear Components (C3)

MLP probes show direction-dependent gains: EN→ZH improves by 3.1 pp (Gemma), consistent with 2.3 pp (Qwen) and ∼4.7 pp bootstrap-corrected (Llama). The weaker transfer direction benefits more from non-

Table 7: Language transition onset and probe peak across architectures.

Model	Transition Onset (%)	Probe Peak (%)	CKA at Peak
Llama-3.1-8B	55%	88%	0.64
Qwen3-8B	50%	67%	0.73
Gemma-2-9B-IT	52%	72%	0.70

Table 8: PCA cluster metrics across layers (Llama-3.1-8B). Low silhouette values reflect the moderate probe accuracy (∼75%), not a clean separation.

Layer	Depth %	Silhouette	EN–ZH Overlap	Centroid Dist.
Layer 10	31%	0.18	0.78	0.82
Layer 18	56%	0.28	0.62	1.24
Layer 25	78%	0.31	0.48	1.73
Layer 28	88%	0.34	0.41	2.08

linear modeling, suggesting that EN→ZH mapping involves more complex cross-lingual transformations. Architectures with stronger multilingual alignment (Qwen, with higher ZH training proportion) show smaller non-linear gains, consistent with simpler cross-lingual mappings.

5.4 Limitations

1. **Two language pairs**: EN–ZH and EN–JA provide suggestive but insufficient evidence for broad typological claims. Additional pairs (EN–DE, EN–AR) would strengthen the linguistic distance analysis.
2. **Dataset**: ConflictQA covers entity-centric factual conflicts; generalization to reasoning-intensive conflicts (where CLEAR found resource abundance dominates) is untested.
3. **Causal evidence**: Steering results are correlational. The 34% flip rate indicates the probe direction *overlaps with* but does not *constitute* the causal mechanism.
4. **SAE analysis**: Base–instruct mismatch rendered Track 2 results inconclusive (Appendix B).
5. **Model scale**: All models are 8–9B parameters; behavior at larger scales may differ.

5.5 Future Directions

Two high-priority directions emerge: (1) SAE analysis with matched training distributions (training SAEs on the instruct models directly or using base models throughout), enabling clean H1-vs-H2 feature-overlap tests; (2) cross-lingual SAE-based steering (Track 3), deferred here due to unreliable feature identification, which would establish whether monolingual conflict features can steer cross-lingual behavior.

6 Conclusion

We presented a systematic investigation of cross-lingual knowledge conflict encoding in multilingual LLMs. Our main finding is that conflict signals are linearly detectable in residual streams and exhibit a remarkably consistent directional asymmetry: transfer from non-English to English consistently exceeds the reverse, with a pooled effect of 6.6 ± 0.9 pp across three architectures ($I^2 = 0\%$). This effect-size stability, rather than individual significance tests, constitutes the primary evidence for an English-anchored conflict encoding mechanism.

Layer-wise CKA analysis reveals that conflict detection peaks during the language-universal-to-specific representation transition, and probe-based steering provides correlational evidence consistent with a causal role (Cohen's $d = 0.41$). Our confound-exclusion pipeline (token overlap, COMET ablation, label noise

Table 9: Steering intervention results (Llama-3.1-8B, best probe layer).

Condition	Flip Rate (%)	Coherence (%)	Random Ctrl (%)
ZH→EN probe direction	34.2	74.1	5.8
EN→ZH probe direction	18.1	71.3	6.2
Averaged probe direction	26.5	72.8	6.0
Random orthogonal direction	6.0	89.5	—

Table 10: Steering strength sensitivity (ZH→EN direction, Llama-3.1-8B).

α	Flip Rate (%)	Coherence (%)	Net Useful Flips (%)
1.0	12.3	91.2	11.2
2.0	22.8	84.6	19.3
3.5	34.2	74.1	25.3
5.0	38.9	58.3	22.7
7.0	41.2	42.1	17.3

robustness) establishes that these results reflect genuine semantic-level signals rather than surface artifacts. We report preliminary SAE results honestly as a boundary of current tools, identifying the base–instruct distribution mismatch as a critical methodological challenge for future mechanistic interpretability work in multilingual settings.

A Label Noise Robustness

To assess sensitivity to verification pipeline errors (estimated at ∼4%), we inject systematic label noise at rates from 5% to 20% (Table 11).

Table 11: Probe accuracy under systematic label noise (Llama-3.1-8B, EN→EN).

Noise Rate	Bal. Acc (%)	Δ (pp)	95% CI
0%	77.1	—	[74.3, 79.9]
5%	75.8	−1.3	[72.9, 78.7]
10%	73.8	−3.3	[70.7, 76.9]
15%	70.4	−6.7	[67.0, 73.8]
20%	66.8	−10.3	[63.1, 70.5]

At the estimated pipeline error rate of 4%, the accuracy loss is ∼1 pp, well within the CI width. Degradation accelerates beyond 10%, but the noise-tolerance threshold (10%) far exceeds our estimated error rate, confirming that label noise does not materially affect conclusions.

B Preliminary SAE Analysis

B.1 Qwen-Scope Conflict Feature Identification

We applied Qwen-Scope pretrained SAEs (65,536 features, Qwen3-8B-Base) to conflict vs. control activations at layer 22. Table 12 summarizes the results.

The base–instruct cosine similarity of 0.71 indicates that ∼30% of activation patterns are misaligned, causing the SAE to decompose instruct-specific patterns into multiple sparse features. Only 35% of MI-selected features are manually interpretable, and the low EN–ZH overlap ($8/45 \approx 18\%$) likely reflects noise rather than genuine language specificity. We caution against interpreting these results as evidence for or against H1/H2.

Table 12: SAE conflict feature statistics (Qwen-Scope, layer 22). Results are noisy due to base–instruct distribution mismatch (cosine similarity 0.71).

Metric	EN	ZH	Overlap
Activated features ($> 2\sigma$)	312	256	78
MI-selected (> 0.15)	45	42	8
Interpretable (manual)	16/45	14/42	3/8
Mean activation diff (conflict−control)	2.31	1.87	—

B.2 LangFIR Language Feature Decomposition

LangFIR random-token filtering identified 120 language-identity features. Of the 45 MI-selected conflict features, 22 (49%) overlap with language-identity features (Table 13). After filtering, 23 residual conflict features remain with only 3 cross-lingual overlaps and 22% interpretability.

Table 13: LangFIR conflict–language feature decomposition.

Category	Count	Cross-lingual Overlap	Interpretable
All MI-selected conflict features	45	8 (18%)	16 (36%)
Language-identity features (LangFIR)	120	—	—
Conflict ∩ Language	22	—	—
Residual conflict (post-filter)	23	3 (13%)	5 (22%)

The high overlap rate (49%) suggests that many "conflict features" are actually language features mislabeled due to upstream SAE noise, further confirming that base–instruct mismatch is the primary limitation.

B.3 Cross-Lingual Feature Jaccard

Table 14 reports cross-lingual Jaccard indices by layer group. Both values are near zero (0.04–0.06) with no significant difference ($p = 0.38$).

Table 14: Cross-lingual conflict feature Jaccard by layer group. Near-zero values reflect measurement noise.

Layer Group	Depth %	EN feat.	ZH feat.	Jaccard	Perm. p
Mid (10–18)	31–56%	38	32	0.06	—
Late (25–30)	78–94%	28	24	0.04	—
Mid − Late diff	—	—	—	+0.02	0.38

This result is *inconclusive* rather than negative: when individual feature identification is unreliable, overlap metrics are necessarily noisy. The direction (mid > late) is consistent with H1 but the difference is not meaningful given the noise floor. SAE analysis with matched training distributions is a critical next step.

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